Also note that

$$H(\delta_{x_1}\mathcal{K},\delta_{x_2}\mathcal{K}) = \sup_{(y_1,y_2)\in\mathbb{Y}^2} (\imath_{x_1,x_2}(\mathcal{K})(y_1)\ \imath_{x_2,x_1}(\mathcal{K})(y_2))$$

with the Radon-Nikodym function $\imath_{x_1,x_2}(\mathcal{K})(y)$ defined in (3.7). We also have

$$\sup_{(x_1,x_2)\in S^2} H(\delta_{x_1}\mathcal{K},\delta_{x_2}\mathcal{K}) = 1/\hbar(\mathcal{K})$$

with the parameter $\hbar(\mathcal{K})$ defined in (3.8). Using the fact that

$$\tanh\left(\frac{1}{4}\ \log u\right) = \frac{u^{1/4}-u^{-1/4}}{u^{1/4}-u^{-1/4}} = \frac{u^{1/2}-1}{u^{1/2}+1} = \frac{1-u^{-1/2}}{1+u^{-1/2}}$$

we conclude that

$$\chi_H(\mathcal{K}) = \frac{1-\sqrt{\hbar(\mathcal{K})}}{1+\sqrt{\hbar(\mathcal{K})}}. \tag{B.2}$$

Proposition B.1. *Consider the collection of Sinkhorn transitions $\mathcal{S}_n$ defined in (2.1). For any $n \geqslant 0$,*

$$\hbar(\mathcal{S}_n) = \hbar(\mathcal{Q}) \quad \text{and} \quad \chi_H(\mathcal{S}_n) = \frac{1-\sqrt{\hbar(\mathcal{Q})}}{1+\sqrt{\hbar(\mathcal{Q})}}.$$

Proof. For any $n \geqslant 0$, we have

$$\begin{aligned}\imath_{y_1,y_2}(\mathcal{S}_{2n+1})(x_1)\ \imath_{y_2,y_1}(\mathcal{S}_{2n+1})(x_2) &= \frac{(d\delta_{x_1}\mathcal{S}_{2n}/d\pi_{2n})(y_1)}{(d\delta_{x_1}\mathcal{S}_{2n}/d\pi_{2n})(y_2)}\frac{(d\delta_{x_2}\mathcal{S}_{2n}/d\pi_{2n})(y_2)}{(d\delta_{x_2}\mathcal{S}_{2n}/d\pi_{2n})(y_1)}\\ &= \imath_{x_2,x_1}(\mathcal{S}_{2n})(y_2)\ \imath_{x_1,x_2}(\mathcal{S}_{2n})(y_1)\end{aligned}$$

which, together with (3.8), yields

$$\hbar(\mathcal{S}_{2n+1}) = \hbar(\mathcal{S}_{2n}).$$

In the same vein, for any $n \geqslant 1$ we have

$$\begin{aligned}\imath_{x_1,x_2}(\mathcal{S}_{2n})(y_1)\ \imath_{x_2,x_1}(\mathcal{S}_{2n})(y_2) &= \frac{(d\delta_{y_1}\mathcal{S}_{2n-1}/d\pi_{2n-1})(x_1)}{(d\delta_{y_1}\mathcal{S}_{2n-1}/d\pi_{2n-1})(x_2)}\frac{(d\delta_{y_2}\mathcal{S}_{2n-1}/d\pi_{2n-1})(x_2)}{(d\delta_{y_2}\mathcal{S}_{2n-1}/d\pi_{2n-1})(x_1)}\\ &= \imath_{y_2,y_1}(\mathcal{S}_{2n-1})(x_2)\ \imath_{y_1,y_2}(\mathcal{S}_{2n-1})(x_1),\end{aligned}$$

hence

$$\hbar(\mathcal{S}_{2n}) = \hbar(\mathcal{S}_{2n-1}).$$

Recalling that $\mathcal{S}_0 = \mathcal{Q}$, this ends the proof of the proposition. ∎

Appendix C. Weighted norm contractions

We associate with some $\varphi \in \mathcal{B}(\mathbb{X})$ and $\psi \in \mathcal{B}(\mathbb{Y})$ such that $\varphi(x) \wedge \psi(y) \geqslant 1$ the following condition.

$\mathcal{L}_{\varphi,\psi}(\mathcal{K})$ *There exists some parameters $\epsilon \in]0,1[$ and $c < \infty$ such that*

$$\mathcal{K}(\psi) \leqslant \epsilon\ \varphi + c. \tag{C.1}$$

In addition, there exist some constant $r_0 \geqslant 1$ and some function

$$\begin{array}{rcccl}\varepsilon & : & [r_0,\infty[& \to &]0,1] \\ & & r & \mapsto & \varepsilon(r)\end{array}$$

such that, for any $r \geqslant r_0$ and $(x_1,x_2) \in C_\varphi(r)^2$,

$$\chi_\varphi(r,\mathcal{K}) := \sup_{(x_1,x_2)\in C_\varphi(r)^2} \|(\delta_{x_1}-\delta_{x_2})\mathcal{K}\|_{\mathsf{tv}} < 1-\varepsilon(r). \tag{C.2}$$

By (3.3), condition (C.2) is equivalent to the existence of some $\mathcal{Y} \in \mathcal{M}_1(\mathbb{Y})$ (that may depends on (r, x_1, x_2)) satisfying for $i \in \{1, 2\}$ the local minorization condition

$$\mathcal{K}(x_i, dy) \geqslant \varepsilon(r)\ \mathcal{Y}(dy). \tag{C.3}$$

For instance, assume that $\delta_{x_1}\mathcal{K} \ll \delta_{x_2}\mathcal{K}$ for any $x_1, x_2 \in \mathbb{X}$ and set

$$\jmath_{\varphi,\psi}(r, \mathcal{K}) := \inf_{(x_1,x_2)\in C_\varphi(r)}\ \inf_{y\in C_\psi(r)} \imath_{x_1,x_2}(\mathcal{K})(y). \tag{C.4}$$

In this notation, we see that if $\jmath_{\varphi,\psi}(r, \mathcal{K}) \geqslant \varepsilon(r)$ then (C.3) holds.

Note that the strength of conditions (C.1), (C.2) depends on the strength of the functions (φ, ψ). When the functions are bounded, the geometric drift condition (C.1) is trivially met, but in this case (C.3) is a uniform contraction condition on the space $\mathbb{X}$.

When $\varphi \in \mathcal{B}_\infty(\mathbb{X})$ the r-sub-level sets $C_\varphi(r)$ are compact and condition (C.3) is met as soon as we have

$$\mathcal{K}(x, dy) \geqslant q(x, y)\ \nu(dy) \tag{C.5}$$

for some Radon positive measure ν on $\mathbb{Y}$ and some density function q, satisfying the local minorization condition

$$\inf_{C_\varphi(r)\times C_\psi(r)} q > 0 \quad \text{and} \quad 0 < \nu(C_\psi(r)) < \infty \tag{C.6}$$

for any $r \geqslant r_0$. For locally compact Polish spaces condition $0 < \nu(C_\psi(r)) < \infty$ is satisfied if ψ has compact sub-levels sets $C_\psi(r)$ with non empty interior and ν is a Radon measure of full support. Also note that the l.h.s. minorization condition (C.6) is satisfied when $(x, y) \in (\mathbb{X}\times\mathbb{Y})^\circ \mapsto q(x, y)$ is a continuous positive function on the interior $(\mathbb{X}\times\mathbb{Y})^\circ := (\mathbb{X}^\circ \times \mathbb{Y}^\circ)$ of $(\mathbb{X}\times\mathbb{Y})$. Also note that

$$\text{if} \quad \mathcal{K}(\psi)/\varphi \leqslant \Theta \in \mathcal{B}_0(\mathbb{X}) \quad \text{then (C.1) holds.} \tag{C.7}$$

Indeed, if the inequality above holds then for any $0 < \epsilon < \|\Theta\|$, the set $A_\epsilon := \{\Theta \geqslant \epsilon\} = C_{1/\Theta}(1/\epsilon)$ is a non empty compact subset and we have

$$\mathcal{K}(\psi) \leqslant \epsilon\ 1_{S-A_\epsilon}\varphi + 1_{A_\epsilon}(\Theta\ \varphi) \leqslant \epsilon\ \varphi + c_\epsilon \quad \text{with} \quad c_\epsilon := \|\Theta\| \sup_{A_\epsilon} \varphi. \tag{C.8}$$

Theorem C.1. *Assume Condition $\mathcal{L}_{\varphi,\psi}(\mathcal{K})$ is satisfied for some uniformly positive functions $(\varphi, \psi) \in (\mathcal{B}(\mathbb{X}) \times \mathcal{B}(\mathbb{Y}))$. Then, there exist constants $\kappa \geqslant 0$ and $\rho \in [0, 1[$ that only depend on the parameters (ϵ, c) in (C.1) and on the function $\varepsilon(r)$ in (C.2) and such that*

$$\chi_{\varphi_\kappa,\psi_\kappa}(\mathcal{K}) \leqslant \rho \quad \textit{with} \quad (\varphi_\kappa, \psi_\kappa) := (1 + \kappa\varphi, 1 + \kappa\psi). \tag{C.9}$$

Theorem C.1 is a consequence of Lemma 2.3 in [41], see also [39], as well as Theorem 8.2.21 in [44] and Lemma 2.1 in [6]. Several examples satisfying the Lyapunov condition (C.1) and the local minorization conditions (C.9) are discussed in [6, 44].

Proof of Theorem 4.6. We check that if Condition $\mathcal{D}_{\varphi,\psi}(\mathcal{S})$ is satisfied for some integer $n_0 \geqslant 0$, then (4.6) and (4.7) also hold for any $n > n_0$.

Assume that Condition $\mathcal{D}_{\varphi,\psi}(\mathcal{S})$ holds for some $n_0 \geqslant 0$. From (4.13), for any $n > n_0$ we have the estimates

$$\begin{aligned} \mathcal{S}^\circ_{2n}(\psi) &= \mathcal{S}_{2n-1}(\mathcal{S}_{2n}(\psi)) \leqslant \epsilon\ \mathcal{S}_{2n-1}(\varphi) + c \leqslant \epsilon^2\ \psi + (1+\epsilon)\ c \quad \text{and} \\ \mathcal{S}^\circ_{2n+1}(\varphi) &= \mathcal{S}_{2n}(\mathcal{S}_{2n+1}(\varphi)) \leqslant \epsilon\ \mathcal{S}_{2n}(\psi) + c \leqslant \epsilon^2\ \varphi + (1+\epsilon)\ c. \end{aligned} \tag{C.10}$$

Also note that for any $n > n_0$ the inequality (4.14) in Condition $\mathcal{D}_{\varphi,\psi}(\mathcal{S})$ implies that

$$\|((\delta_{y_1} - \delta_{y_2})\mathcal{S}_{2n-1})\mathcal{S}_{2n}\|_{\mathsf{tv}} \vee \|((\delta_{x_1} - \delta_{x_2})\mathcal{S}_{2n})\mathcal{S}_{2n+1}\|_{\mathsf{tv}} \leqslant 1 - \varepsilon(r). \tag{C.11}$$

The estimates in (C.10) imply that (4.6) holds, while the inequality (C.11) implies that (4.7) is satisfied.

Moreover, combining the fixed point equations in (2.8) with (C.10) we arrive at

$$\sup_{n\geqslant n_0}\left(\|\mathcal{S}_{2n}(\psi)\|_\varphi \vee \|\mathcal{S}_{2n+1}(\varphi)\|_\psi\right) < \infty \quad \text{and} \quad \left(\lambda_U(\varphi)\vee\nu_V(\psi)\right) \leqslant \frac{c}{1-\epsilon}, \tag{C.12}$$

and (4.13) together with (C.12) yields

$$\sup_{n\geqslant n_0}\left(\pi_{2n}(\psi)\vee\pi_{2n+1}(\varphi)\right) \leqslant c_1\ \left(\lambda_U(\varphi)\vee\nu_V(\psi)\right)$$

for some $c_1 > 0$. Therefore, Condition $\mathcal{D}_{\varphi,\psi}(\mathcal{S}^\circ)$ is satisfied.

Finally, we observe that (4.13) ensures that inequality (C.1) is satisfied for the Markov transitions $\mathcal{S}_{2n}$ and $\mathcal{S}_{2n+1}$, while (4.14) yields

$$\chi_\varphi(r,\mathcal{S}_{2n})\vee\chi_\psi(r,\mathcal{S}_{2n+1}) \leqslant 1-\varepsilon(r)$$

and, hence, the inequality (C.2) also holds for $\mathcal{S}_{2n}$ and $\mathcal{S}_{2n+1}$. As a consequence, Theorem C.1 can be applied to show that there are parameters $\kappa\geqslant 0$ and $\rho\in[0,1[$ such that

$$\chi_{\varphi_\kappa,\psi_\kappa}(\mathcal{S}_{2n})\vee\chi_{\psi_\kappa,\varphi_\kappa}(\mathcal{S}_{2n+1}) \leqslant \rho.$$

■

Appendix D. Sinkhorn potential functions

This appendix collects a number of technical results concerning the Sinkhorn potential functions associated with the alternating Sinkhorn dynamics. We first establish a set of commutation formulae linking the successive marginals and Radon–Nikodym derivatives along the iteration. These identities are then used to derive quantitative bounds on the Sinkhorn potentials and on the corresponding density ratios, both in uniform and weighted norms. Under suitable integrability and Lyapunov-type assumptions, we further obtain refined estimates leading to drift and local minorization inequalities for the Sinkhorn transition kernels. Together, these results provide the analytical backbone needed to verify the contraction and stability properties used in the proofs of the main theorems.

D.1. **Commutation formulae.** For any $n\geqslant 0$ we have the commutation formulae

$$\mathcal{S}_{2n}\left(\frac{d\nu_V}{d\pi_{2n}}\right) = \frac{d\pi_{2n+1}}{d\lambda_U} \quad \text{and} \quad \mathcal{S}_{2(n+1)}\left(\frac{d\pi_{2n}}{d\nu_V}\right) = \frac{d\lambda_U}{d\pi_{2n+1}}. \tag{D.1}$$

In the same vein, for any $n\geqslant 1$ we have

$$\mathcal{S}_{2n-1}\left(\frac{d\lambda_U}{d\pi_{2n-1}}\right) = \frac{d\pi_{2n}}{d\nu_V} \quad \text{and} \quad \mathcal{S}_{2n+1}\left(\frac{d\pi_{2n-1}}{d\lambda_U}\right) = \frac{d\nu_V}{d\pi_{2n}}. \tag{D.2}$$

A proof for (D.1) and (D.2) is provided in Appendix E (on page 47).

Using the commutation properties (D.1) and (D.2) for any $n\geqslant p$ we readily check that

$$\begin{aligned}\frac{d\pi_{2n}}{d\nu_V} &= \mathcal{S}^\circ_{2n}\left(\frac{d\pi_{2(n-1)}}{d\nu_V}\right) = \mathcal{S}^\circ_{2n,2p}\left(\frac{d\pi_{2p}}{d\nu_V}\right) \quad \text{and}\\ \frac{d\pi_{2n+1}}{d\lambda_U} &= \mathcal{S}^\circ_{2n+1}\left(\frac{d\pi_{2n-1}}{d\lambda_U}\right) = \mathcal{S}^\circ_{2n+1,2p-1}\left(\frac{d\pi_{2p-1}}{d\lambda_U}\right),\end{aligned} \tag{D.3}$$

with the backward semigroups

$$\mathcal{S}^\circ_{2n,2p} := \mathcal{S}^\circ_{2n}\ldots\mathcal{S}^\circ_{2(p+1)} \quad \text{and} \quad \mathcal{S}^\circ_{2n+1,2p-1} := \mathcal{S}^\circ_{2n+1}\ldots\mathcal{S}^\circ_{2p+1}.$$